

Graduation Project Documentation

Impact of Short Video Consumption on Individual Behavior and Daily Interactions

Project Idea

This project aims to analyze the impact of short-form video content consumption on individuals' behavior, productivity, attention span, social interactions, and daily habits. The study is based on survey data collected from participants with different age groups and usage patterns.

Problem Statement

With the rapid growth of short-video platforms such as TikTok, YouTube Shorts, and Instagram Reels, people spend increasing amounts of time consuming fast-paced digital content. However, the behavioral and social effects of this type of content consumption remain unclear. The project analyzes whether excessive exposure to short videos negatively affects focus, productivity, social communication, and daily decision-making.

Project Objectives

- Analyze user behavior related to short-video consumption.
- Identify correlations between screen time and behavioral changes.
- Discover patterns and trends using data analysis techniques.
- Build dashboards and visual reports.
- Generate actionable insights and recommendations.

Tools & Technologies

- Microsoft Excel → Basic data cleaning and preprocessing.
- SQL → Data extraction, querying, and management.
- Python → Deep data cleaning, preprocessing, exploratory data analysis (EDA), and visualization.
- Tableau → Interactive dashboards and visual storytelling.
- Power BI → Business intelligence dashboards and reporting.

Project Planning & Management

The project is developed collaboratively, where all team members contribute equally across all project phases instead of dividing fixed responsibilities.

The team jointly participates in:

- Data Cleaning
- Deep Cleaning using Python
- Data Transformation
- SQL Querying
- Exploratory Data Analysis (EDA)
- Dashboard Design
- Data Visualization
- Result Interpretation
- Documentation Preparation

This collaborative workflow ensures equal practical exposure and skill development for all team members.

Literature Review

Recent studies indicate that short-form digital content significantly affects users' attention spans, cognitive engagement, and social behavior. Previous research has discussed digital addiction, attention fragmentation, reduced productivity, and social isolation.

However, many previous studies focused mainly on theoretical discussions rather than practical data analysis workflows. This project addresses that gap by applying modern data analytics tools to real survey data.

Research Gap

Most previous studies discuss psychological and behavioral impacts theoretically without applying real-world analytical workflows on collected datasets. This project bridges that gap by using real survey data and applying data cleaning, SQL analysis, Python analysis, and BI visualization tools to generate measurable insights.

Requirements Gathering

Functional Requirements:

- Import and clean survey datasets.
- Perform deep cleaning using Python.
- Store and query data using SQL.
- Analyze behavioral patterns.
- Generate dashboards and reports.
- Compare different user groups.

Non-Functional Requirements:

- Data accuracy.

- Dashboard clarity.
- Fast query performance.
- Data consistency.
- Easy interaction with dashboards.

System Analysis & Design

System Workflow:

1. Collect survey responses.
2. Clean and preprocess data using Excel and Python.
3. Apply deep cleaning techniques using Python.
4. Store structured data using SQL.
5. Perform exploratory data analysis.
6. Build dashboards using Tableau and Power BI.
7. Generate insights and recommendations.

Expected Outcomes

The project is expected to reveal behavioral trends related to short-video usage, identify the most affected user groups, measure the relationship between screen time and productivity, and provide data-driven recommendations for healthier digital habits.

Data Cleaning, Transformation & Dashboard Development

After collecting the survey data, we translated it from Arabic to English to ensure consistency and better compatibility with data analysis and visualization tools.

We refined the dataset by removing unnecessary formatting such as brackets and asterisks, which helped standardize the responses and improve data quality.

Using Python, we performed comprehensive data cleaning processes including handling missing values (nulls), removing duplicates, validating data consistency, and standardizing categorical values.

We also split complex and multi-select responses (such as platform categories) into structured and analyzable formats. Additional preprocessing steps included trimming spaces, unifying text formats, and detecting anomalies.

After completing the cleaning and preprocessing phase, we successfully utilized Python to build interactive dashboards, enabling us to visualize insights, identify patterns, and present the data in a clear and impactful way.

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4	أخبار وأح 🌍 😄 **أي وقت فاضي (constantly)** *
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5	أحياناً بأجل سماعة لوقت ثاني
6	بسمعه بسرعه العاديه ومعنديش مشكلة
7	معنديش خلق اسمع فويسات فا بسمعه x2
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